

Introduction to Qualitative Methodology Part One

Presentation By Dr Teresa Burdett

Learning Outcomes

- By the end of this lecture, students will be able to:
- Distinguish between ontology, epistemology, methodology, and methods.
- Explain how philosophical assumptions inform research design in clinical research.
- Compare major research paradigms (positivism, post-positivism, interpretivism, critical paradigms).
- Identify strengths and limitations of different research designs used in clinical/health
- research.
- Describe foundational principles of qualitative research, including sampling, data collection, and analysis.

Clinical Research

Key definitions



Ontology – what we believe about reality



Epistemology – what counts as knowledge and how we know it



Methodology – the strategic plan based on a philosophical stance



Methods – the tools used to collect/analyse data

Why this matters in clinical research?



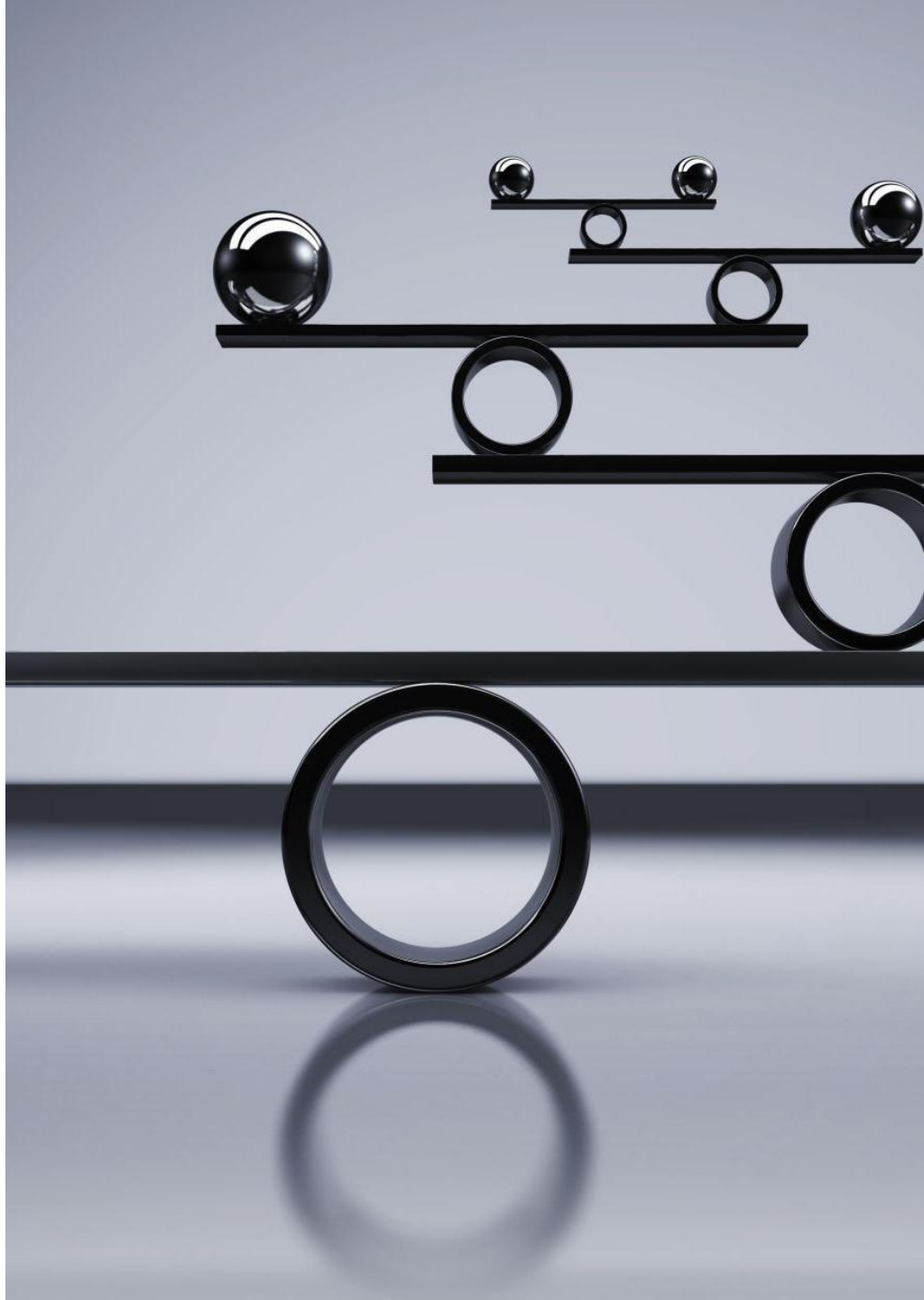
Determines the relevance of methods used



Shapes decisions on sample sizes, validity, and interpretability of findings

Research Paradigms

- Overview or philosophical framework guiding a study
- Reality (ontology), knowledge (epistemology) and values (axiology)
- Positivism vs interpretivism



Ontology



Ontological Assumptions

- Ontology is a branch of philosophy that examines the nature of reality and
- existence.
- In research, ontology explores:
  - What exists?
  - What is real?
  - How do entities relate to each other?

Importance of Ontology

- Shapes research paradigms and methodologies.
- Determines how researchers perceive the world and approach knowledge construction.

Ontology – Realism

Reality exists independently of human thoughts, perceptions, or beliefs.

Truth and facts are absolute and discoverable through scientific methods.

Key Features:

- Reality is external and does not change based on beliefs.
- Associated with positivism and quantitative research.

Ontology – Relativism

- Reality is subjective and shaped by human experiences, beliefs, and culture.
- No single truth—reality varies across individuals and societies.
- Key Features:
  - Knowledge is shaped by context and social interactions.
  - Associated with interpretivism and qualitative research

Ontology – Pragmatism

Reality is fluid and defined by what works best in a given situation.

Rejects strict divisions between objective and subjective reality.

Key Features:

- Reality is problem-oriented and changeable.
- Associated with mixed-methods research (qualitative + quantitative).

Ontology – Critical Realism (Constructivism)

- Reality is real but influenced by historical, political, and social structures.
- Bridges realism and relativism—reality exists, but perception is shaped by power structures.
- Key Features:
  - Focuses on underlying mechanisms and hidden social structures.
  - Used in social sciences, critical theory, and political studies.

…but subjectivism

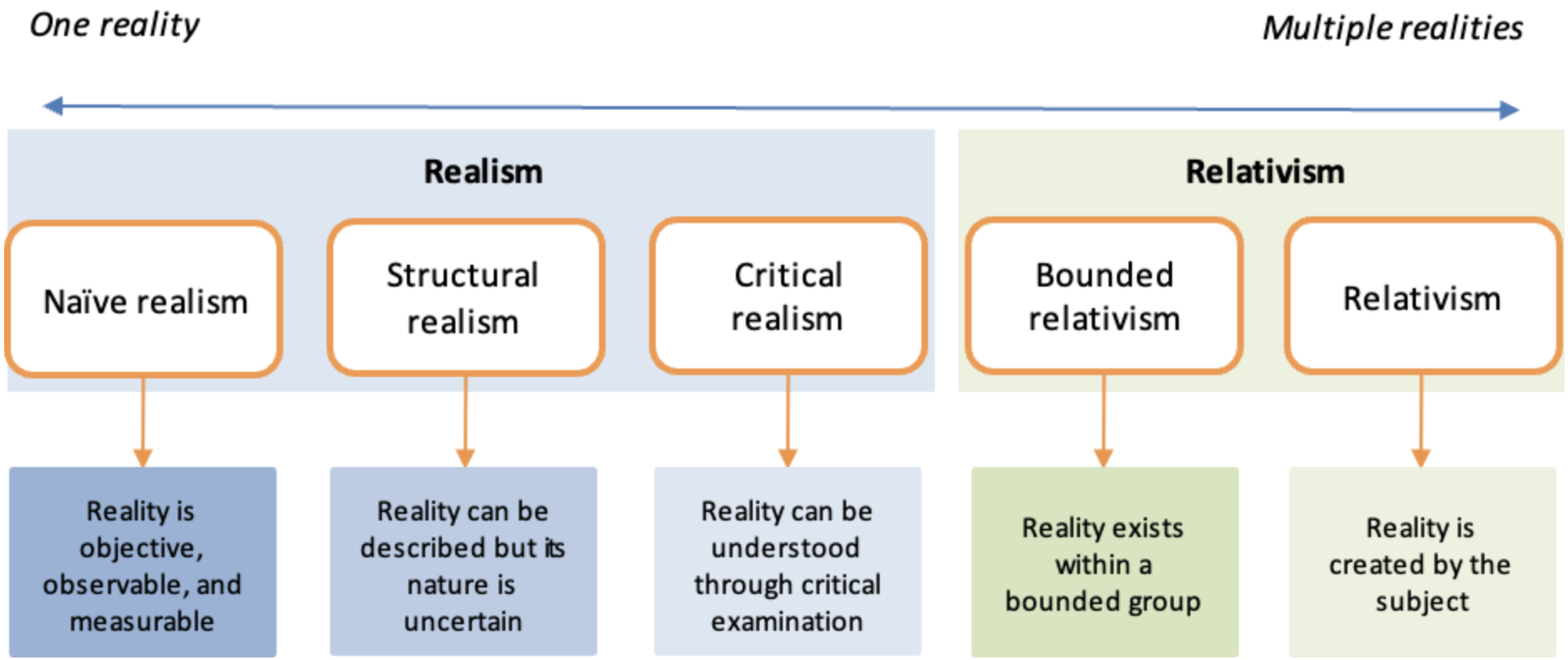
Reality is created by individuals

Reality is entirely shaped by individual consciousness, and there is no existence beyond personal experiences.

Each person creates their own reality based on thoughts, emotions, and beliefs.

Used in phenomenological and narrative research, where researchers explore personal stories and emotions.

So Ontology is therefore…





Why Ontology Matters in Research?

- Shapes research questions
- Guides method selection
- Provides a lens
- Informs your paradigm

Ontology Clinical Research Examples

Ontology	Example in Clinical Research
Realism	RCT assumes objective measurable outcomes.
Critical Realism	Mixed-methods evaluation of a nutrition intervention.
Relativism	Interview study exploring patient lived experiences of IBS.

Epistemology

Epistemology



Epistemology is the branch of philosophy that studies:

The nature of knowledge

How knowledge is acquired

What counts as valid evidence



Key Question:

"How do we know what we know?"

Importance of Epistemology

- Determines what methods are appropriate for collecting and analysing data.
- Influences the choice of quantitative, qualitative, or mixed-methods approaches.
- Hennink, Hutter, & Bailey (2018)
  - "Epistemology refers to what counts as knowledge and evidence."

Core Concepts in Research Epistemology



Nature of knowledge



Sources of knowledge



Justification

Epistemology – Positivism/ Objectivism

- Positivism assumes that knowledge is objective, measurable, and independent of human perception.
- Reality consists of facts that can be observed and quantified using scientific methods.
- Key features
  - Objectivity
  - Quantitative methods
  - Causality

Epistemology – Interpretivism/ Constructivism

- Knowledge is constructed through human experiences, interactions, and cultural context.
- Unlike positivism, interpretivism does not seek objective facts but rather aims to understand how individuals perceive and experience reality.
- Key features
  - Lived experiences
  - Qualitative methods
  - Researcher involvement
  - No absolute truth

Comparison of Positivism vs. Interpretivism

Epistemological Approach	Positivism	Interpretivism
Nature of Knowledge	Objective, measurable facts	Subjective, socially constructed meanings
Reality Perception	Exists independently of human interpretation	Depends on human experience and context
Methods Used	Quantitative (surveys, experiments, statistical analysis)	Qualitative (interviews, ethnography, observations)
Researcher's Role	Detached, neutral observer	Active participant in meaning-making
Focus	Explaining and predicting	Understanding and interpreting
Context Consideration	Minimal (focus on universal laws)	High (meaning shaped by culture, history, social factors)

Choosing the Right Epistemological Approach

Use Positivism when:

✓ Seeking objective, measurable data.
✓ Using experiments, surveys, or statistical methods.
✓ Aiming to establish causal relationships.
✓ Example: Studying the impact of new medication on blood pressure.

Use Interpretivism when:

✓ Seeking in-depth understanding of human experiences.
✓ Using interviews, case studies, ethnography.
✓ Exploring social or cultural influences on behavior.
✓ Example: Investigating how patients emotionally cope with chronic illness.

Choosing the right approach ensures research validity and relevance.

Impact on Methodology

- **Epistemological Stance**
- The epistemological stance forms the foundation of research methodology, guiding the
- researcher's approach to inquiry.

- **Method Selection**
- The choice of methods for data collection is directly influenced by the underlying epistemological stance of the research.

- **Data Analysis Techniques**
- Data analysis techniques are shaped by the chosen methods and epistemological stance, impacting research outcomes.

Research designs in clinical research – Qualitative Research

Definition of Qualitative Research



Understand “how” and “why”



Behaviours, emotions, experiences and beliefs

"Qualitative research consists of any kind of research that produces findings not arrived at by means of statistical procedures or other means of quantification. Instead, it focuses on the in-depth understanding of human experiences and social contexts.“ (**Patton, 2015)**:

Reference: Patton, M. Q. (2015). Qualitative research & evaluation methods: Integrating theory and practice (4th ed.). SAGE Publications.

Qualitative Research – Key Features



Focuses on understanding human behaviour and experiences.



Uses open-ended, exploratory questions.



Collects data in natural settings (e.g., interviews, observations).



Emphasizes meaning, context, and depth over numerical data.



Involves subjective interpretation by researchers.

When to Use Qualitative Research

- Exploring new topics
- Understanding cultural and social contexts
- Examining personal experiences and narratives
- Analysing group dynamics and interactions
- Studying change over time
- Investigating language and media
- When quantitative methods are insufficient
- Conducting participatory or action research



Key Differences Between Qualitative and Quantitative Research

Aspect	Qualitative Research	Quantitative Research
Nature of Data	Non-numerical (texts, images, videos, observations)	Numerical (statistics, percentages, figures)
Objective	Explores in-depth meanings and patterns	Tests hypotheses and measures relationships
Approach	Subjective, interpretative	Objective, empirical
Data Collection Methods	Interviews, observations, focus groups, case studies, document analysis	Surveys, experiments, structured observations, secondary data
Analysis Methods	Thematic analysis, content analysis, discourse analysis	Statistical tests (t-tests, chi-square, regression analysis)
Research Design	Flexible and evolving during the study	Structured and follows a fixed plan
Sample Size	Small, purposive, non-random	Large, random, representative
Generalizability	Limited to specific contexts	High generalizability to larger populations
Researcher's Role	Active participant in interpretation	Detached, independent observer
Example Research Questions	"How do students experience online learning?"	"What percentage of students prefer online learning over in-person classes?"

Major Qualitative Methodologies

- Phenomenology
- Ethnography
- Grounded Theory
- Case Study
- Narrative Research
- Action Research
- Discourse Analysis

Ethnography

- Definition:
  - Immersing in a cultural or social group to study behaviours, norms, and interactions in natural settings.
- Key Features
  - Long-term fieldwork with participant observation.
  - Unstructured interviews to gain authentic insights.
  - Thick descriptions capturing detailed social dynamics.



Phenomenology

- Definition:
  - Understanding lived experiences and how individuals perceive a phenomenon.
- Key Features:
  - In-depth interviews to explore subjective experiences.
  - Bracketing to reduce researcher bias.



Grounded Theory

- Definition:
  - Developing theories from systematically collected data.
- Key Features:
  - Inductive approach – theories emerge from data.
  - Constant comparison – data is analysed throughout.
  - Open, axial, and selective coding used in analysis.



Case Study

- Definition:
  - In-depth analysis of a single case or small group within a real-life context.
- Key Features:
  - Uses multiple data sources (e.g., interviews, documents).
  - Can be exploratory, explanatory, or descriptive.



Narrative Research

- Definition:
  - Collecting and analysing life stories and personal experiences.
- Key Features:
  - Chronological or thematic storytelling analysis.



Action Research

- Definition:
  - A collaborative approach where researchers and participants work together to solve a problem.
- Key Features:
  - Cyclical process: Planning → Action → Observation → Reflection.
  - Stakeholder involvement for real-world impact.



Discourse Research

- Definition:
  - Examining spoken and written language in social contexts to understand meaning and power structures.
- Key Features:
  - Analyses texts, conversations, and media.
  - Focuses on social structures, identity, and power relations



Data Collection Methods

- Observations.
- Document/ Content Analysis
- In-Depth Interviews
- Focus Group Discussions (FGDs)
- Visual Methods
- Life Stories & Biographies

Data Analysis



Thematic Analysis

Content Analysis



Narrative Analysis



Discourse Analysis

Key Characteristics



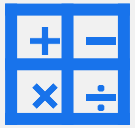
Naturalistic Inquiry



Subjectivity and Interpretive Analysis



Inductive Approach



Use of Non-Numerical Data

Sampling Strategies

- Flexibility and Adaptability
- Holistic Perspective
- Small and Purposeful Samples
- Researcher as the Primary Instrument

Data Analysis, Findings, and Ethics



Emphasis On Meaning And Context



Multiple Data Collection Methods



Iterative Data Analysis



Rich Descriptive Findings

Advantages & Challenges of Qualitative Research

Advantages:

- Provides rich, detailed insights.
- Captures social and cultural contexts.
- Allows flexibility in exploring emerging ideas.

Challenges:

- Time-consuming and labour-intensive.
- Difficult to generalize findings.
- Requires careful interpretation to minimize researcher bias

Paradigms of Qualitative Research

Research Paradigms

A research paradigm is a philosophical framework that guides how research is conducted.



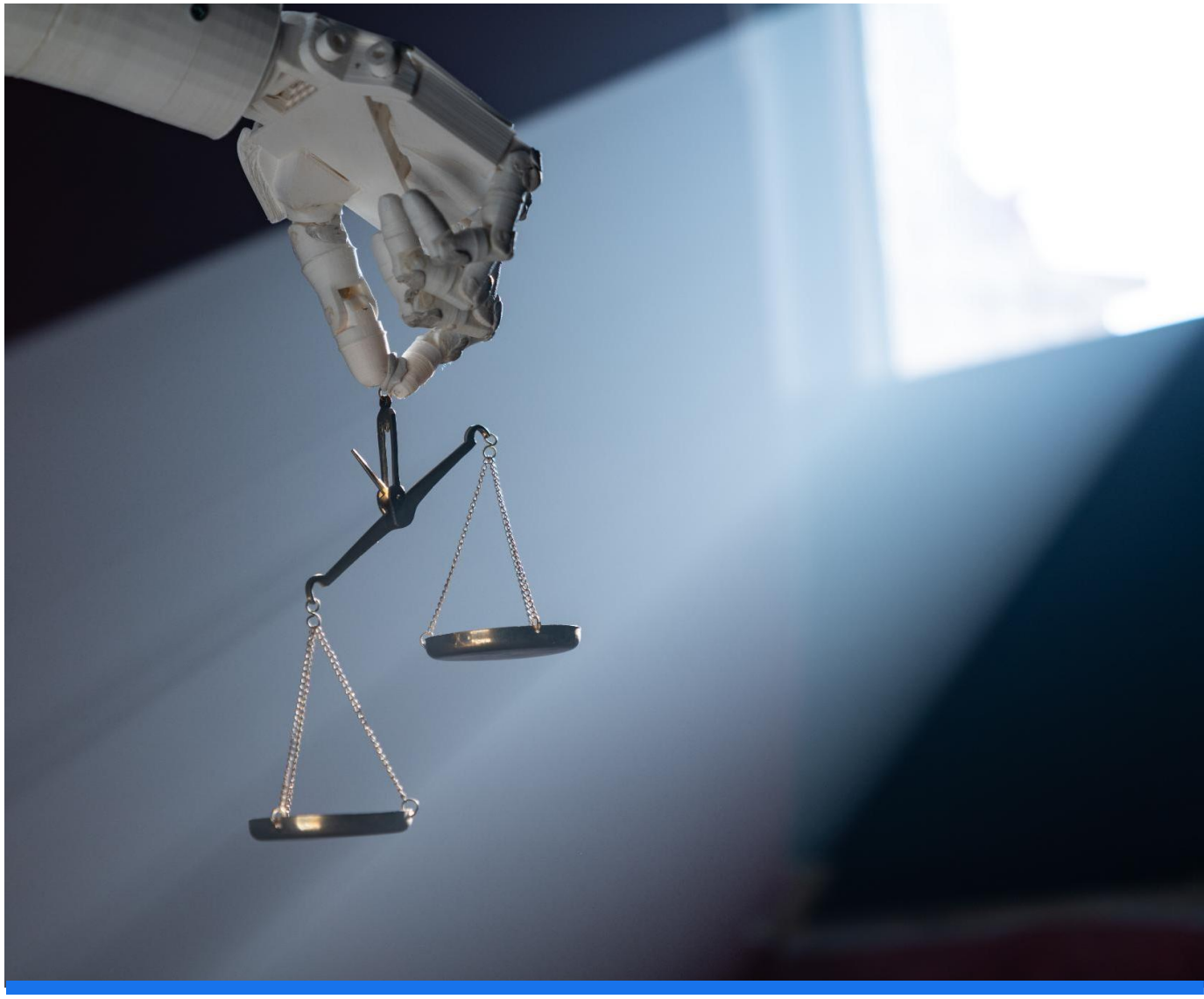
Types of Research Paradigms

- Major paradigms in qualitative research include:
  - Hermeneutics (Interpretation Approach)
  - Ethnography (Understanding Culture & Behaviour)
  - Phenomenology (Lived Experiences)
  - Symbolic Interactionism (Meaning through Symbols)



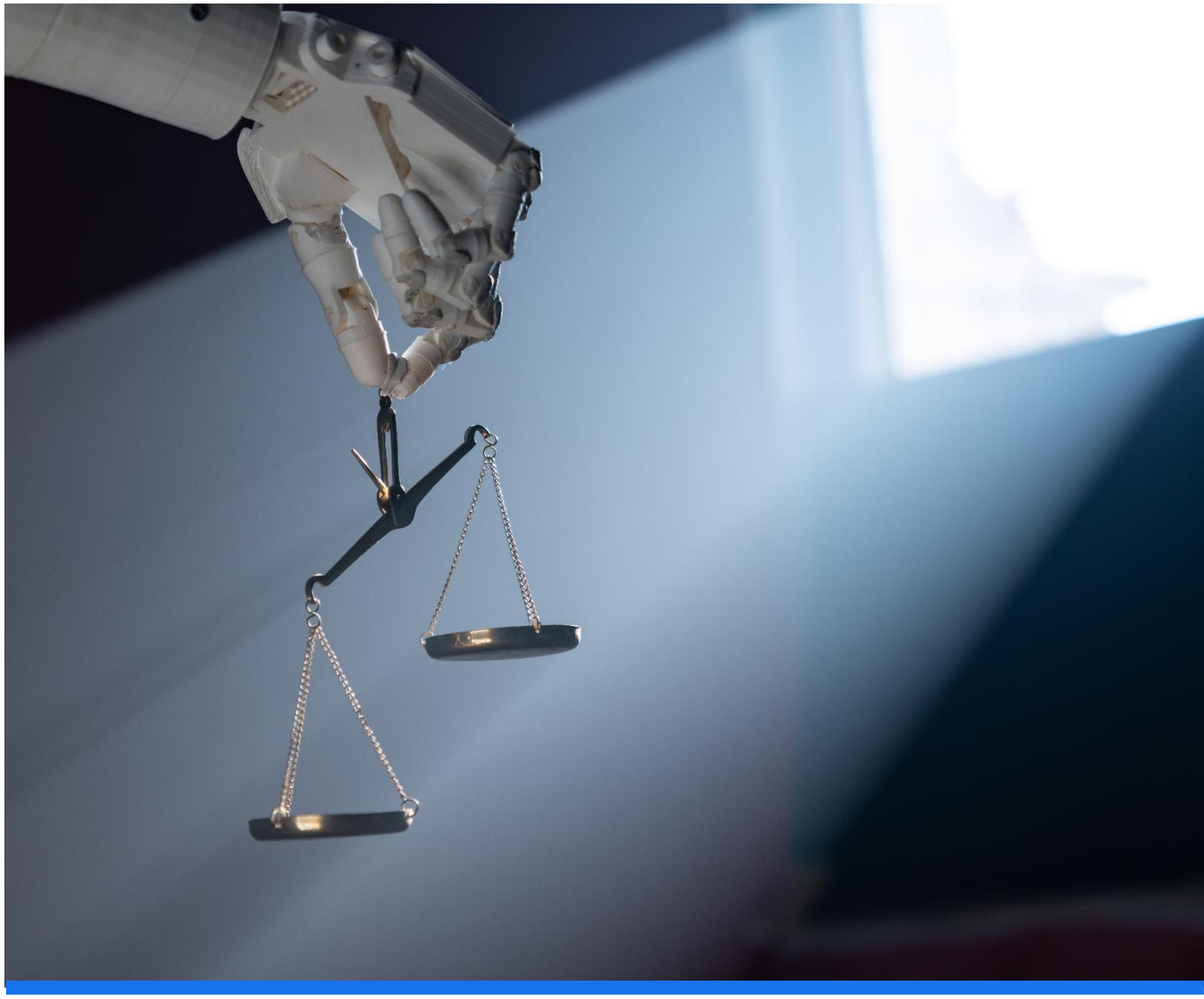
Hermeneutics (Interpretation Approach)

- Study of interpretation of texts, symbols, and human experiences.
- Meaning is contextual, historical, and subjective.
- Key Characteristics
  - Focuses on interpreting texts, interviews, narratives, historical documents.
  - Encourages reflexivity (researcher awareness of biases).



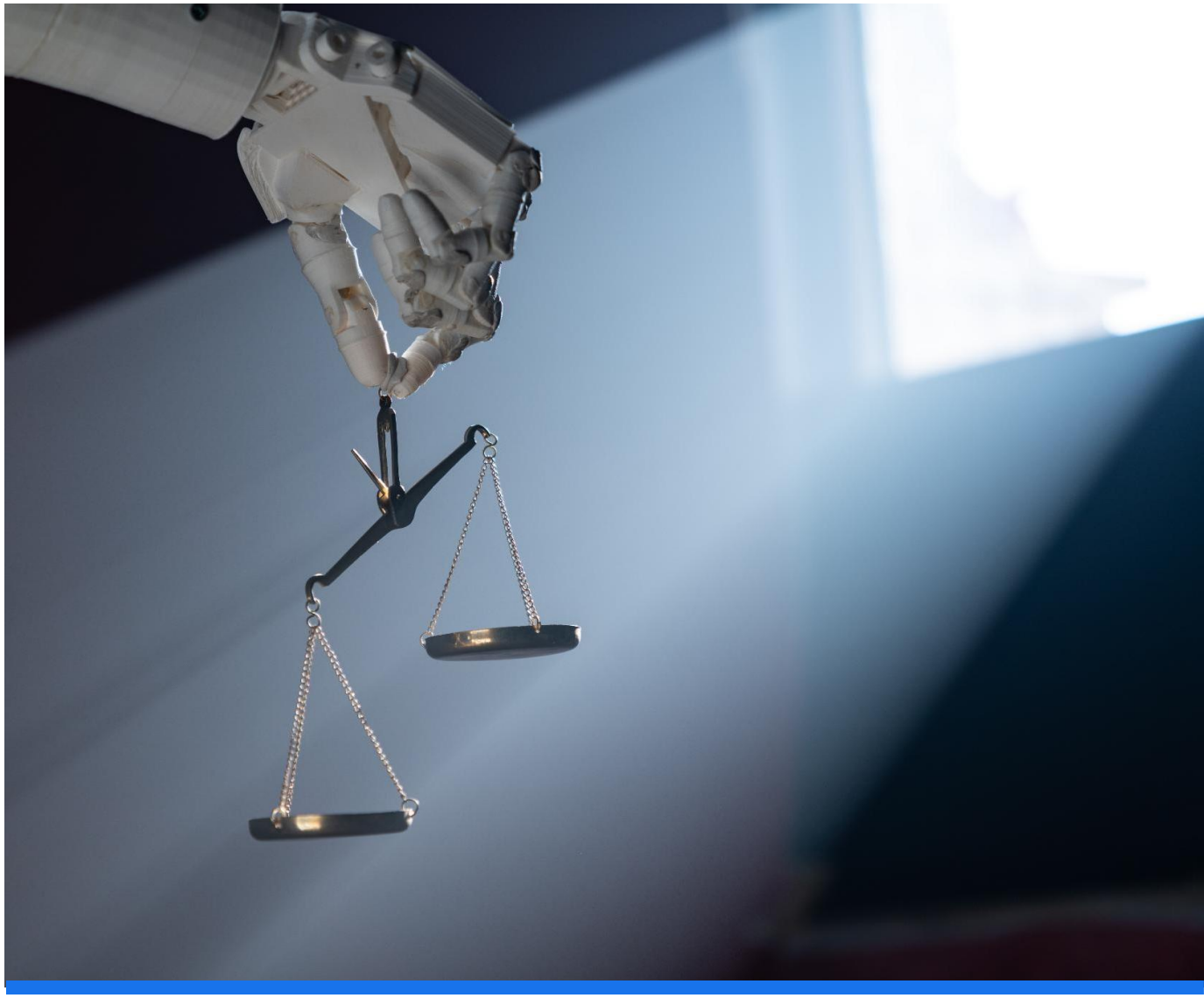
Ethnography (Understanding through Experiences)

- Study of cultures, behaviours, and social practices through immersion in a community.
- Key Characteristics:
  - Conducted through participant observation, interviews, and field notes.
  - Uses thick descriptions (detailed cultural accounts).



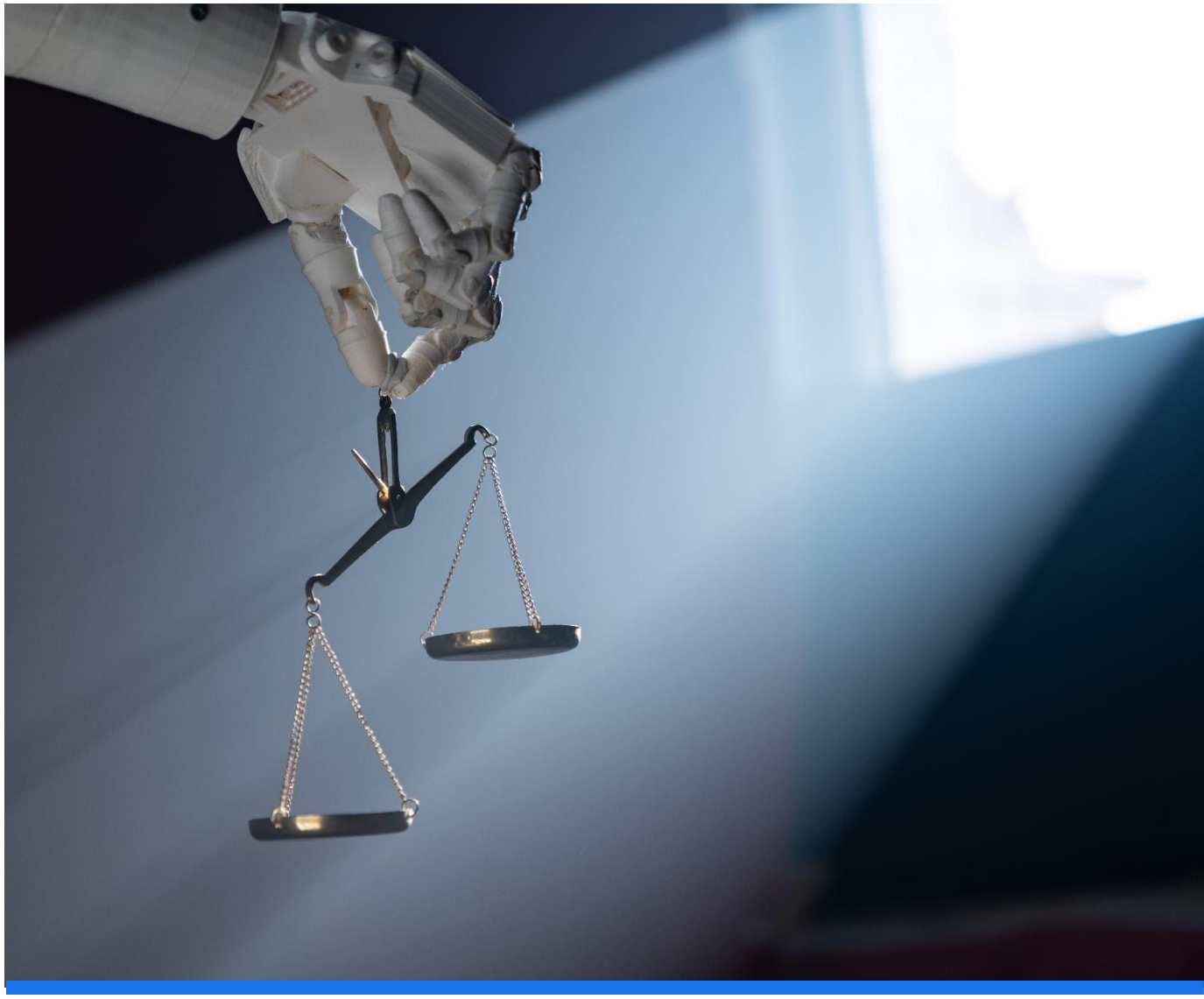
Phenomenology (Lived Experiences)

- Focuses on how individuals experience and interpret the world.
- Key Characteristics
  - Uses in-depth interviews and personal narratives.
  - Involves bracketing (suspending researcher biases).
  - Examines subjective meanings of lived experiences.



Symbolic Interactionism (Meaning Through Symbols)

- Examines how people create meaning through language, gestures, and social cues.
- Key Characteristics:
  - Meaning is socially constructed through interactions.
  - Symbols are interpreted based on social context.
  - Focuses on micro-level interactions (e.g., conversations, identity formation).



References

- Clandinin, D. J., & Connelly, F. M. (2000). *Narrative inquiry: Experience and story in qualitative research*. Jossey-Bass.
- Geertz, C. (1973). *The interpretation of cultures: Selected essays*. Basic Books.
- Hammersley, M., & Atkinson, P. (2019). *Ethnography: Principles in practice* (4th ed.). Routledge.
- Riessman, C. K. (2008). *Narrative methods for the human sciences*. SAGE Publications.
- Van Manen, M. (2016). *Researching lived experience: Human science for an action sensitive pedagogy* (2nd ed.). Routledge.
- Yin, R. K. (2018). *Case study research and applications: Design and methods* (6th ed.). SAGE Publications.

References

- Gee, J. P. (2014). *An introduction to discourse analysis: Theory and method* (4th ed.). Routledge.
- Gray, E. (2022). Doing Research In The Real World (5th ed.) Sage Publications.
- Kemmis, S., McTaggart, R., & Nixon, R. (2013). *The action research planner: Doing critical participatory action research*. Springer.
- Krueger, R. A., & Casey, M. A. (2015). *Focus groups: A practical guide for applied research* (5th ed.). SAGE Publications.